EXP NO:13 EXPERIMENT TO HANDLE MISSING VALUES AND INAPPROPRIATE DATA IN A GIVEN DATA SET

CODE :

```
import numpy as np
import pandas as pd
df = pd.read_csv("Hotel_Dataset.csv")
df.drop_duplicates(inplace=True)
df.reset_index(drop=True, inplace=True)
df.loc[df.CustomerID < 0, 'CustomerID'] = np.nan
df.loc[df.Bill < 0, 'Bill'] = np.nan
df.loc[df.EstimatedSalary < 0, 'EstimatedSalary'] = np.nan
df.loc[(df['NoOfPax'] < 1) | (df['NoOfPax'] > 20), 'NoOfPax'] = np.nan
df["Hotel"] = df["Hotel"].replace("Ibys", "Ibis")
df["FoodPreference"] = df["FoodPreference"].replace(
   {"Vegetarian": "Veg", "veg": "Veg", "non-Veg": "Non-Veg"}
)
df["EstimatedSalary"] =
df["EstimatedSalary"].fillna(round(df["EstimatedSalary"].mean()))
df["NoOfPax"] = df["NoOfPax"].fillna(round(df["NoOfPax"].median()))
df["Rating(1-5)"] = df["Rating(1-5)"].fillna(round(df["Rating(1-
5)"].median()))
df["Bill"] = df["Bill"].fillna(round(df["Bill"].mean()))
print(df)
```

OUTPUT :

C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_5300\2580639570.py:1: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

df.CustomerID.loc[df.CustomerID<0]=np.nan

df.Bill.loc[df.Bill<0]=np.nan

C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_5300\2580639570.py:2: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame

df.EstimatedSalary.loc[df.EstimatedSalary<0]=np.nan

C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_5300\2129877948.py:1: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

df['NoOfPax'].loc[(df['NoOfPax']<1) | (df['NoOfPax']>20)]=np.nanarray(['20-25', '30-35', '25-30', '35+'], dtype=object)

array(['Ibis', 'LemonTree', 'RedFox', 'Ibys'], dtype=object)

<bound method Series.unique of 0 veg

1 Non-Veg

2 Veg

3 Veg

4 Vegetarian

5 Non-Veg

6 Vegetarian

7 Veg

8 Non-Veg

9 non-Veg

Name: FoodPreference, dtype: object>

	CustomerID	Age_Group	Hotel	FoodPreference	NoOfPax	Bill	\
0	1.0	30-35	Ibis	Non-Veg	5.0	9227.0	
1	2.0	35+	LemonTree	Non-Veg	3.0	6203.0	
2	3.0	20-25	RedFox	Veg	6.0	2798.0	

3	4.0	30-35	RedFox	Non-Veg	2.0	6291.0
4	5.0	30-35	RedFox	Veg	3.0	6495.0
..	...	...	...	...	...	...
595	596.0	35+	RedFox	Veg	7.0	7530.0
596	597.0	20-25	Ibis	Veg	7.0	1258.0
597	598.0	30-35	Ibis	Veg	8.0	1282.0
598	599.0	35+	LemonTree	Veg	1.0	2898.0
599	600.0	25-30	LemonTree	Non-Veg	8.0	5135.0

	EstimatedSalary	Rating(1-5)
0	26629.0	1
1	85031.0	1
2	53380.0	1
3	37071.0	5
4	40074.0	5
..	...	...
595	90263.0	2
596	73117.0	1
597	78394.0	1
598	43782.0	2
599	48454.0	5

[600 rows x 8 columns]